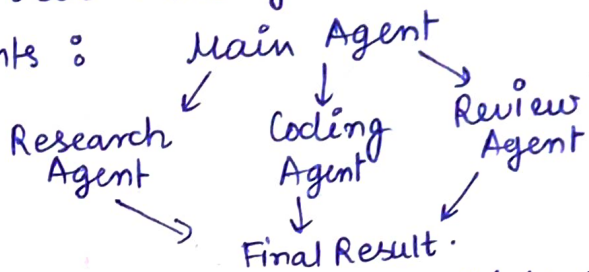
09. Agent-to-Agent COMMUNICATION

Agent-to-Agent (A2A) communication

→ AI agents exchange tasks, information, context, states & results to collaborate.

Instead of one agent: User → One Agent → Final Answer.

Multiple Specialised Agents:



Main Idea: Multiple agents collaborate to complete a larger goal.

Why Multiple agents?

Complex tasks can be divided among specialized agents.

Research Agent → Analysis Agent → Writing Agent → Review Agent.

Benefit: Specializations + Collaboration.

Agent Roles

Research Agent → collects info
Analysis " → analyzes info
Coding " → handles code
Review Agent → checks work
Coordinator " → assigns tasks & combines results.

Delegation *

→ assigning a subtask to another agent.

Main agent → Research task → Research Agent → Research Result → Main Agent

(#) you do this subtask & give me the result.

Handsoff *

→ transferring responsibility of a task to another agent.

⊙ customer support agent → Billing Issue → Billing Agent → Handles the issue

⊙ Delegation: Do a subtask

⊙ Handoff: take over responsibility.

Agent Communication

Agents exchange messages containing:

• Task • Context • Input • Instructions • States • Result • Erro

Basic flow : Agent A → Send Task / Message → Agent B →
→ Process Task → Send Result → Agent A.

Shared Context & State

Shared context → relevant info used by multiple agents
State → current progress / status of the task.

Eg : Task ⇒ Process Refund
customer_verified = true order_found = true refund_eligible = true
refund_processed = false # context Engineering (connection).

Synchronous Communication *

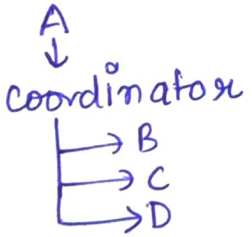
Sender waits for the response
Agent A → Request → Agent B → Response → Agent A continues.
useful when the next step depends on the response.

Asynchronous Communication *

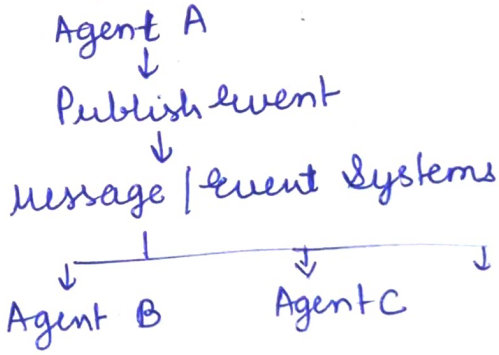
Sender doesn't wait for an immediate response
Agent A → Send task → Continues other work → Agent B finishes →
→ Result available
useful for independent tasks.

Communication patterns *

⊙ Request-Response : Agent A → Req → Agent B → Response.
⊙ Sequential : A → B → C
⊙ Parallel :

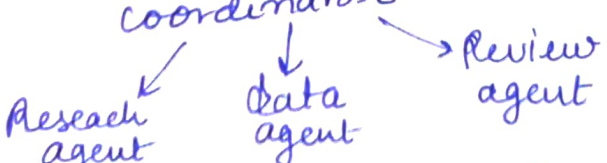


⊙ Publish / Subscribe :



Coordinator Pattern

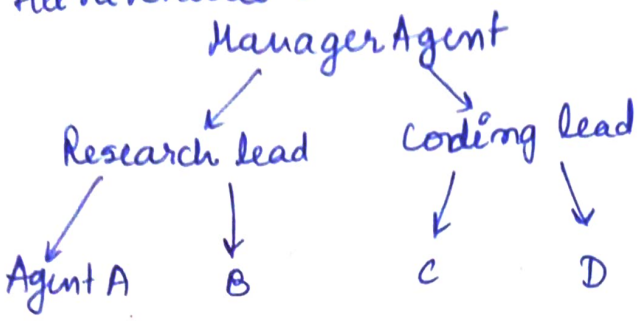
→ coordinator Agent manages other agents.



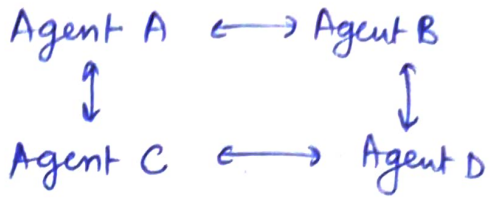
Eg : breaks task → assigns subtasks → tracks progress →
collects results → combines outputs. # AI orchestration (connection).

Hierarchical v/s Peer-to-Peer

Hierarchical :



Peer-to-Peer :



agents communicate directly without a central coordinator.

Structured communication

Good messages should clearly specify :
Task + Input / context + Expected output.

Poor : "Check this candidate"

Better : Task = Evaluate internship eligibility.
Input = Candidate Profile + internship requirements.
Output = Eligible / Not eligible + reasons.

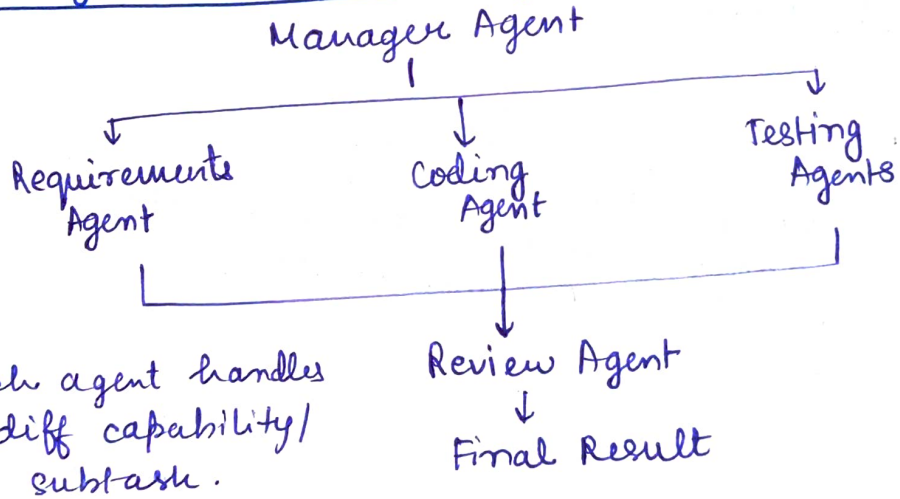
structured communication reduces misunderstanding.

Context Passing

Pass relevant context to the next agent.
Main Agent → Relevant Context → Research Agent.

Don't pass unnecessary information.

Multi-Agent Problem Solving



Each agent handles a diff capability / subtask.

Error Handling & Validation

Possible problems : → Agent doesn't respond
→ invalid / wrong format → Missing information.
→ contradictory result → Misunderstood task
→ Repeated requests

Basic handling : Invalid Response → Validate → Retry | Clarify | Fallback.

Never assume another agent's output is automatically correct.

Prevent Infinite loops

Bad : Agent A → Agent B Agent B → Agent A
Agent A → Agent B Agent B → Agent A ---

Use : maximum interactions • timeout • termination condition • task completion check.

A2A v/s Tool Calling

Tool calling : Agent → DB Agent → Search API Agent → Calculator

A2A communication : Agent A → Agent B

A tool performs an operation ; another agent can perform its own reasoning | task.

Orchestration v/s A2A

Orchestration → controls the overall workflow.
A2A communication → allow agents in that workflow to exchange tasks, information, context and results.

ORCHESTRATION → controls workflow → Agent A → Agent B → Agent C
← A2A communication.

Must Remember

- A2A → agents communicate & callobrate.
- Use structured msgs + relevant content
- Validate outputs + handle failures + prevent loops.
- Orchestration coordinates, A2A communicates.